

Group 7 Final Report

Stats M148 FingerHut Project

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1 Introduction

context and background info. cite any external sources

- definition of diff journey statuses

2 Exploratory Data Analysis

2.1 Plots:

- table of journeys (successful, unsuccessful, incomplete)
- distribution of journey durations split by end result
-

3 Task 1: Predictive Modeling of Ongoing Journeys

We modeled whether an ongoing customer journey would eventually end in an `order_shipped` event. To make completed journeys resemble the ongoing journeys seen at prediction time, we cut each completed journey at sampled intermediate points and calculated features using only the actions visible before each cut. We used an XGBoost classifier because its boosted trees can represent nonlinear effects and interactions among event counts, journey timing, and the customer’s most recent activity.

To limit data leakage, features that directly reveal the outcome or occur very late in the purchase process were excluded. These included `n_order_shipped`, order-placement counts, downpayment counts, and the original action timestamps. We also downsampled successful journeys in training to better reflect their lower prevalence among ongoing journeys. XGBoost performed well and handled the large, sparse feature set, although its predictions are less directly interpretable than those from a regression model and may be sensitive to changes in customer behavior over time.

3.1 XGBoost Model Interpretation

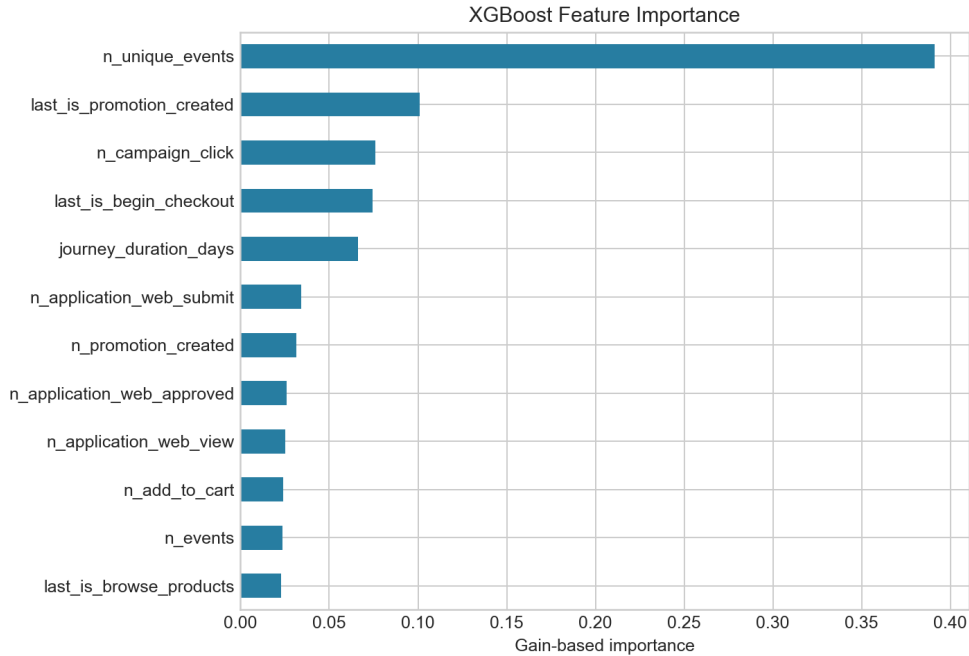


Figure 1: Gain-based importance for the leading XGBoost features. Importance identifies features used heavily by the model, but does not show the direction of their effects.

The number of unique event types observed in a journey was the most important non-leaky feature, so we examined it using individual conditional expectation (ICE) and ceteris paribus (CP) profiles. Both methods vary this feature while holding all other journey features fixed.

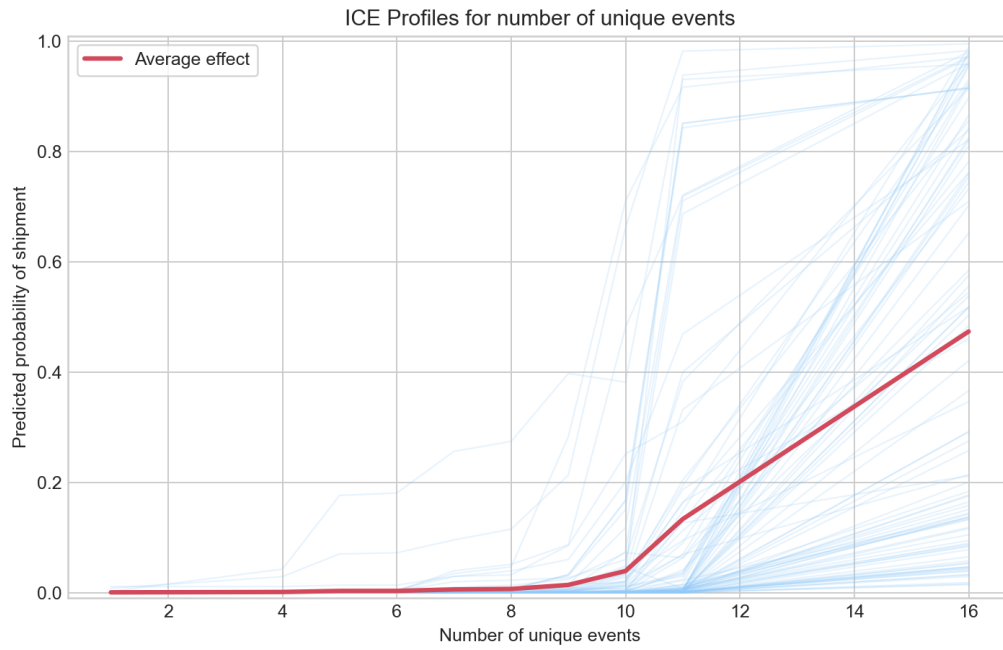


Figure 2: ICE profiles for 100 journeys. Each blue line represents one journey and the red line is their average.

The ICE profiles show a nonlinear effect: predicted shipment probabilities are generally low for journeys with fewer than about ten unique event types, then rise as journeys include a broader range of actions. The large spread among the blue curves shows that event variety is not sufficient by itself; the effect depends strongly on the journey's other events and timing features.

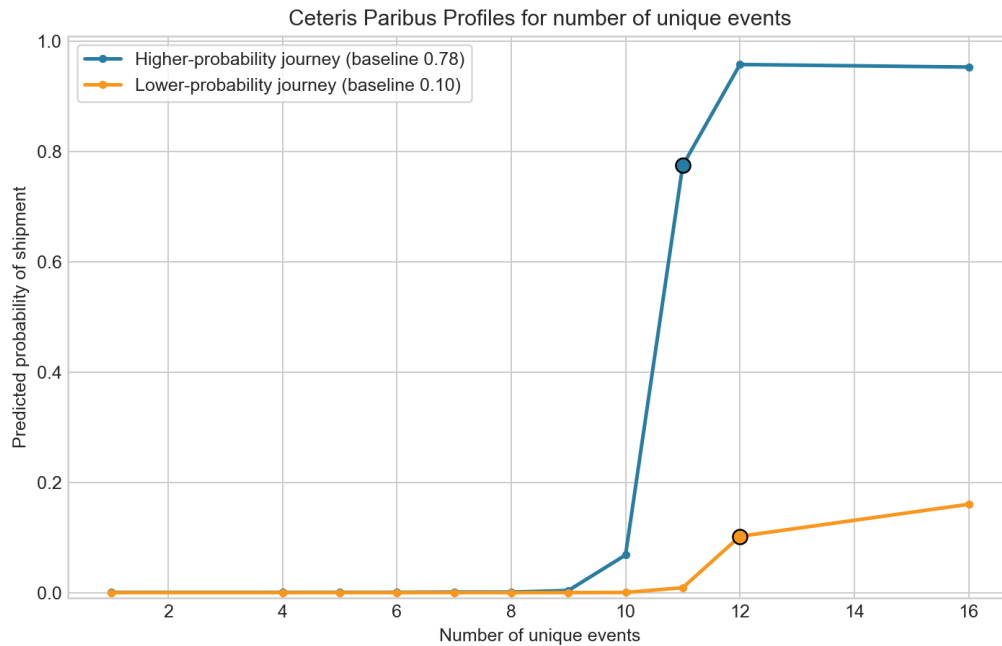


Figure 3: CP profiles for one higher-probability and one lower-probability journey. The outlined points mark each journey's observed value.

The CP profiles provide local explanations for two specific journeys. Increasing event variety sharply raises the higher-probability journey's prediction, while the lower-probability journey remains relatively unlikely to ship even at high values. This difference illustrates the interactions captured by XGBoost: otherwise favorable journey signals must also be present for event variety to translate into a high predicted probability. These profiles describe model behavior rather than a causal effect, and some hypothetical feature combinations may not occur naturally.

4 Task 2: Forecasting Number of Orders Shipped

– task description – modeling approach. Discussion of strengths and weaknesses of the model, possible improvements. – actual forecasts including a quantification of how good your forecasts are – at least 2 visualizations of your model, more is better

- can reuse prophet plots
- need to calculate accuracy by comparing forecast of model trained on `dat_train1` to the new data in `dat_train2`

5 Sidequest: LSTM?

- briefly discuss process and package used, pretty much just copy off slides *at least one more section on something you did in the "branching out" category*

6 Appendix

6.1 Contributions of each team member

- **Charlie:** data cleaning, EDA, XGB tuning + fitting, prophet forecasting, LSTM
- **Isaiah:** data cleaning, EDA, XGB tuning + fitting, prophet forecasting
- **Joel:** data cleaning, EDA, XGB tuning + fitting, prophet forecasting

Contributions of AI: brainstorming, learning NN package for fitting LSTM, ...